SECURE-BLOCK

Milestone 4 Report

Police Reporting System - Advanced Features

This report outlines the technical implementation and features completed for Milestone 4, focusing heavily on real-time data visualization and interactive systems.

1. Live Geographical Tracking System

• Successfully engineered an interactive, open-source mapping system directly into the Citizen Tracking portal.
• Integrated Leaflet and react-leaflet to render high-performance geographical tiles using OpenStreetMap data.
• Configured custom map markers, including a bespoke Police unit icon and Incident Location pins.

2. Real-Time Movement Simulation Engine

• Developed a React use Effect-based interval engine that visually simulates a police patrol car moving toward an active incident.
• Implemented a mathematical interpolation algorithm that smoothly updates the police coordinates every 3 seconds, mimicking real driving approaches.
• Added dynamic rendering logic so the police car only appears if the investigation status is actively "Assigned" or "Under Investigation".

3. Deterministic Coordinate Hashing

• Created an algorithmic string-hashing function that converts the alphanumeric Complaint Tracking ID into a deterministic set of latitude and longitude coordinates.
• Ensures that tracking the same complaint repeatedly will always yield the exact same physical map location without requiring external geocoding API keys.